Solutions to "Algebraic Geometry and Arithmetic Curves"

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1 Some topics in commutative algebra

1.1 Tensor products

Exercise 1.1.1. Let $\{M_i\}_{i\in I}, \{N_j\}_{j\in J}$ be two families of modules over a ring A. Then

$$(\oplus_{i\in I} M_i) \otimes_A (\oplus_{j\in J} N_j) \cong \oplus_{(i,j)\in I\times J}(M_i \otimes_A N_j).$$

Proof. There is a chain of natural isomorphisms:

$$\begin{aligned}
&(\oplus_{i\in I} M_i) \otimes_A (\oplus_{j\in J} N_j)\\
&\cong \oplus_{i\in I} (M_i \otimes_A (\oplus_{j\in J} N_j))\\
&\cong \oplus_{j\in J} \oplus_{i\in I} (M_i \otimes_A N_j)\\
&\cong \oplus_{(i,j)\in I\times J} (M_i \otimes_A N_j).
\end{aligned}$$

□

Exercise 1.1.2. There is a unique A-linear map

$$f : \mathrm{Hom}_A(M, M') \otimes_A \mathrm{Hom}_A(N, N') \to \mathrm{Hom}_A(M \otimes_A N, M' \otimes_A N')$$

such that $f(u \otimes v) = u \otimes v$.

Proof. Consider the map $g : \mathrm{Hom}_A(M, M') \times \mathrm{Hom}_A(N, N') \to \mathrm{Hom}_A(M \otimes_A N, M' \otimes_A N')$ sending (u, v) to $u \otimes v$. It is clearly bilinear, so factors uniquely through $\mathrm{Hom}_A(M, M') \otimes_A \mathrm{Hom}_A(N, N')$. □

Exercise 1.1.3. Let M, N be A-modules and let $i : M' \to M, j : N' \to N$ be submodules of M and N, respectively. Then there is a canonical isomorphism

$$(M/M') \otimes_A (N/N') \cong (M \otimes_A N)/(\mathrm{Im}\, i_N + \mathrm{Im}\, j_N).$$

In particular, $(\mathbf{Z}/n\mathbf{Z}) \otimes_{\mathbf{Z}} (\mathbf{Z}/m\mathbf{Z}) = \mathbf{Z}/l\mathbf{Z}$, where $l = \gcd(m, n)$.

Proof. Observe that

$$(M \otimes_A N)/(\operatorname{Im} i_N + \operatorname{Im} j_N) \cong ((M \otimes_A N)/(\operatorname{Im} i_N))/(\operatorname{Im} j_{M/M'}),$$

so Corollary 1.13 gives us a chain of isomorphisms

$$\begin{aligned} &((M \otimes_A N)/(\operatorname{Im} i_N))/(\operatorname{Im} j_{M/M'}) \\ \cong &((M/M') \otimes_A N)/(\operatorname{Im} j_{M/M'}) \\ \cong &(M/M') \otimes_A (N/N'). \end{aligned}$$

For the last statement, just recall that $(m) + (n) = (l)$. □

Exercise 1.1.4. Let M, N be A-modules and let B, C be A-algebras.

(a) If M and N are finitely generated over A, then so is $M \otimes_A N$.

(b) If B and C are finitely generated over A, then so is $B \otimes_A C$.

(c) The converses of (a) and (b) are in general false.

Proof. Parts (a) and (b) are obvious. For part (c), just notice that $\mathbf{Q} \otimes_{\mathbf{Z}} \mathbf{Z}/2\mathbf{Z}$ is the zero ring, but $\mathbf{Q}$ is not finitely generated over $\mathbf{Z}$. □

Exercise 1.1.5. Let $\rho : A \to B$ be a ring homomorphism, M an A-module, and N a B-module. There is a canonical isomorphism of A-modules

$$\operatorname{Hom}_A(M, \rho_* N) \cong \operatorname{Hom}_B(\rho^* M, N).$$

Proof. Let $f : \operatorname{Hom}_A(M, \rho_* N) \to \operatorname{Hom}_B(\rho^* M, N)$ be the map sending u to the map $f(u)$ s.t. $f(u)(m \otimes b) = bu(m)$. This is well-defined because if $n \otimes c = m \otimes b$, we may assume that there is some $a \in A$ s.t. $an = m$ and $\rho(a)b = c$ and so $f(u)(n \otimes c) = cu(n) = \rho(a)bu(n) = bu(m)$.

In addition, define $g : \operatorname{Hom}_B(\rho^* M, N) \to \operatorname{Hom}_A(M, \rho_* N)$ to be the map s.t. $g(u)(m) = u(m \otimes 1)$. This is immediately seen to be well-defined, and we contest that this is an inverse for f. Indeed, $f(g(u))(m \otimes b) = bg(u)(m) = bu(m \otimes 1) = u(m \otimes b)$ and $g(f(u))(m) = f(u)(m \otimes 1) = u(m)$. □

Exercise 1.1.6. Let $(N_i)_{i \in I}$ be a direct system of modules. Then for any A-module M, there exists a canonical isomorphism $\varinjlim(N_i \otimes_A M) \cong (\varinjlim N_i) \otimes_A M$.

Proof. Tensor-hom.

Just kidding, it follows from the fact that $\varinjlim(N_i \times M)$ is canonically isomorphic to $(\varinjlim N_i) \times M$, and that a bilinear map over the latter must then induce a cocone of bilinear maps. □

Exercise 1.1.7. Let B be an A-algebra, and let M, N be B-modules. Then there exists a canonical surjection $M \otimes_A N \to M \otimes_B N$.

Proof. Do I really need to spell it out? □

1.2 Flatness

Exercise 1.2.1. Let M be an A-module and let $I \subseteq \mathrm{Ann}(m)$ be an ideal.

(a) There is a natural A/I-module structure on M, and $M \cong M \otimes_A A/I$.

(b) Let N be another A-module such that $I \subseteq \mathrm{Ann}(N)$. The canonical homomorphism $M \otimes_A N \to M \otimes_{A/I} N$ is an isomorphism.

Proof. For (a), observe that if $x \in I$, then $am = (a+x)m$ for any $a \in A$ and $m \in M$. This implies that the multiplication of elements of M by cosets $a + I$ is well defined, hence we have an A/I-module structure on M. To see that this is the extension of scalars, recall that $M \otimes_A A/I \cong M/(IM)$, which is just M.

Part (b) follows from a chain of isomorphisms

$$\begin{aligned} & M \otimes_A N \\ \cong & (M \otimes_A A/I) \otimes_{A/I} N \\ \cong & M \otimes_{A/I} N, \end{aligned}$$

where the first isomorphism comes from Corollary 1.11 and the second comes from part (a). □

Exercise 1.2.2. Let $\rho : A \to B$ be a ring homomorphism, S a multiplicative subset of A, and $T = \rho(S)$. Then T is a multiplicative subset of B, and $T^{-1}B \cong B \otimes_A S^{-1}A$.

Proof. The first part is obvious. To see the last statement, let $\iota : B \to T^{-1}B$ be the canonical embedding. Since ι sends elements of $T = \rho(S)$ to units, there is a canonical homomorphism $\phi : S^{-1}A \to T^{-1}B$. Since $\otimes_A$ is a pushout in the category of A-algebras, there is a unique homomorphism $B \otimes_A S^{-1}A \to T^{-1}B$ such that this diagram commutes:

$$\begin{tikzcd} S^{-1}A \otimes B \arrow[dr, dotted] & S^{-1}A \arrow[l] \arrow[d] \\ B \arrow[u] \arrow[r] & T^{-1}B. \end{tikzcd}$$

This is an isomorphism because of universal property of localizations. □

Exercise 1.2.3. Nakayama's lemma does not hold for modules that are not finitely generated.

Proof. Consider $\mathbf{Q}$ as a module over $\mathbf{Z}_{(p)}$. □

Exercise 1.2.4. Let I be an ideal of a ring A. Then the following are equivalent:

(a) A/I is flat over A.

(b) $I^2 = I$

(c) There is an element $e \in I$ such that $e^2 = e$ and $eA = I$.

Proof. ((i)⇒(ii)) Consider the exact sequence $0 \to I \to A \to A/I \to 0$. Since A/I is flat, we have another exact sequence $0 \to A/I \to A/I^2 \to 0$, so $I = I^2$.

((ii)⇒(iii)) I don't know how to do this morally. The easy way out is to use the form of Nakayama's Lemma proven by the determinant trick.

((iii)⇒(i)) Observe that in any maximal ideal, I either localizes to (0) or (1). In any case, it's flat at every maximal ideal, hence flat overall.

□

1.3 Formal completion

Exercise 1.3.1. The usual topology on $\mathbf{R}$ is not given by a subgroup filtration.

Proof. Use noggin for ≥ 2 seconds. □

2 General properties of schemes

2.1 Spectrum of a ring

Exercise 2.1.1. For k a field, $\mathrm{Spec}(k[[T]])$ is the space $\{\xi, 0\}$ with generic point ξ and closed point 0.

Proof. It is easy to see that any power series with a non-zero constant term is a unit. Furthermore, if we have $f = \sum a_n T^n$ with $a_N = 1$ for some N and $a_n = 0$ whenever $n \leq N$, then we can pull out a common factor of T^N. We see that f is then a unit multiple of T^N and $(f) = (T^N)$. Clearly, this is only prime when $N = 1$. It is easy to see that $k[[T]]$ is an entire ring, so the proposition is immediate. □

Exercise 2.1.2. Let $\varphi : A \to B$ be a homomorphism of finitely generated algebras over a field k. The image of a closed point under $\mathrm{Spec}\,\varphi$ is a closed point.

Proof. If $\mathrm{Spec}\,\varphi$ takes the maximal ideal $\mathfrak{m}$ into the ideal $\mathfrak{p}$, we get a diagram

$$k \to A/\mathfrak{p} \xrightarrow{\widetilde{\varphi}} B/\mathfrak{m}.$$

By Corollary 1.12, $B/\mathfrak{m}$ is a finite algebraic extension of k, so $B/\mathfrak{m}$ (resp. $A/\mathfrak{p}$) is finite over $A/\mathfrak{p}$ (resp. k). It then suffices to prove that $A/\mathfrak{p}$ is a field. Any $a \in A$ has an associated irreducible polynomial $P = \sum a_i X^i \in k[x]$ such that $P(a) = 0$. This polynomial must have a nonzero constant term, so we have

$$-\frac{1}{a_0} \sum_{i>0} a_i a^i = 1,$$

but the polynomial on the left has no constant term and we can pull out a factor of a. □

Exercise 2.1.3. Let $k = \mathbf{R}$ be the field of real numbers and let $A = k[X,Y]/(X^2 + Y^2 + 1)$. Let x, y be the respective images of X, Y in A.

(a) Let $\mathfrak{m}$ be a maximal ideal of A. Then there are $a, b, c, d \in k$ such that $x^2 + ax + b, y^2 + cy + d \in \mathfrak{m}$. Furhtermore, $\mathfrak{m}$ is generated by a single element $f = \alpha x + \beta y + \gamma$ with at least one of α, β nonzero.

(b) The map $(\alpha, \beta, \gamma) \mapsto (\alpha x + \beta y + \gamma)A$ establishes a bijection between the subset $\mathbf{P}(k^3) \setminus \{(0,0,1)\}$ of the projective space $\mathbf{P}(k^3)$ and the set of maximal ideals of A.

(c) Let $\mathfrak{p}$ be a non-maximal prime ideal of A. Then the canonical homomorphism $\varphi : k[X] \to A$ is finite and injective, and $k[X] \cap \mathfrak{p} = 0$. Let $g \in \mathfrak{p}$ and let $g^n + a_{n-1}g^{n-1} + \ldots + a_0 = 0$ be an integral equation with $a_i \in k[X]$. Then $a_0 = 0$, and so $\mathfrak{p} = 0$.

Proof. First, notice that $A/\mathfrak{m}$ is isomorphic to $\mathbf{C}$, since $X^2 + Y^2 + 1$ does not split in k. Then the existence of a, b, c, d follows from the fact that any complex number satisfies a monic quadratic equation. Since $\mathbf{C}$ is degree 2 over k, there are α, β, γ, not all zero, such that

$$\alpha x' + \beta y' + \gamma = 0,$$

where x' and y' are the respective images of x and y. Since not all of them are zero, we may assume that α is nonzero. Then we get an equation for x' in terms of y', which we can then substitute into the equation $x'^2 + y'^2 + 1 = 0$. This new equation cannot split in k because that would imply a real root of $x'^2 + y'^2 + 1 = 0$, so we have that $\mathfrak{m} = fA$, where f is the ideal corresponding to the linear equation in x' and y'.

Part (b) holds, since a solution to $X^2 + Y^2 + 1 = 0$ is determined (up to conjugacy) by an arbitrary complex number, which itself determines a linear equation with real coefficients.

That φ is finite and injective is clear, so consider the induced homomorphism $\widetilde{\varphi} : k[X]/\varphi^{-1}(\mathfrak{p}) \to A/\mathfrak{p}$. Notice that $B = k[X]/\varphi^{-1}(\mathfrak{p})$ is not a field, since $\widetilde{\varphi}$ is still finite and injective, which implies that $A/\mathfrak{p}$ is a field as well. Then $B = k[X]$ and $\mathfrak{p} \cap k[X] = 0$. Finally, in the integral equation for g, notice that a_0 must be a multiple of g, hence 0. Then the equation has a common factor of g which we may cancel, which gives a new equation with constant term a_1. This repeats indefinitely and so $g = 0$. □

Exercise 2.1.4. Let A be a ring.

(a) Let $\mathfrak{p}$ be a minimal prime ideal of A. Then every element of $\mathfrak{p}A_{\mathfrak{p}}$ is nilpotent. In particular, every element of $\mathfrak{p}$ is a zero divisor.

(b) If A is reduced, then any zero divisor in A is an element of a minimal prime ideal. This is not true in general if A is not reduced.

Proof. Part (a) is obvious.

For (b), let x be any zero divisor and let $\mathfrak{p}$ be a prime ideal minimal among those that contain x. If $\mathfrak{p}$ is minimal, we are done. Otherwise, suppose that $\mathfrak{q} \subset \mathfrak{p}$ is a minimal prime ideal not containing x. Since A is reduced, we must have $\sqrt{(x)} \cap \mathrm{Ann}(x) = \{0\}$, so $\mathfrak{q}$ must contain a nonzero element $x' \in \mathrm{Ann}(x)$. Localizing by $\mathfrak{q}$, we see that x is sent to 0 and x' is sent to a nilpotent. This implies that there is some positive integer n and some element $s \in A \setminus \mathfrak{q}$ such that $s(x'^n - x) = 0$, and it is easy to see that this means that $sx^2 = 0 \in \mathfrak{q}$, so $x^2 \in \mathfrak{q}$, contradiction. □

The fact that this is false for A not reduced comes is immediate from the fact that there can be non-nilpotent zero divisors in a ring.

Exercise 2.1.5. Let k be a field. Let $\mathfrak{m}$ be a maximal ideal of $k[T_1, \dots, T_n]$.

(a) Let $P_1(T_1)$ be a generator of $\mathfrak{m} \cap k[T_1]$ and $k_1 = k[T_1]/(P_1)$. Then we have an exact sequence

$$0 \to P_1 k[T_1, \dots, T_n] \to k[T_1, \dots, T_n] \to k_1[T_2, \dots, T_n] \to 0.$$

(b) There exist n polynomials $P_1(T_1), P_2(T_1, T_2), \dots, P_n(T_1, \dots, T_n)$ such that $k[T_1, \dots, T_i] \cap \mathfrak{m} = (P_1, \dots, P_i)$ for all $i \leq n$. In particular, $\mathfrak{m}$ is generated by n elements.

Proof. Part (a) is obvious. In part (b), we proceed by induction. The statement is obvious in the single-variable case, so suppose that it holds for $n - 1$, with n an integer at least 2. Let k_1 be as in (a) and observe that it is a field, so the hypotheses and the exact sequence from (a) give a generating set $(P_2, \dots, P_n)$ for the image of $\mathfrak{m}$ in $k_1[T_2, \dots, T_n]$. This pulls back to a set of generators $(P_1, \dots, P_n)$ for $\mathfrak{m}$, and it is easy to see that the desired properties hold. □

Exercise 2.1.6. Any closed subspace of a quasicompact space is itself quasicompact. If A is a ring, $\mathrm{Spec}(A)$ is quasicompact.

Proof. Let X be a topological space and let $Y \subseteq X$ be any closed subset. If $\{U_i\}$ is an open covering of Y, $\{U_i \cup \complement_X Y\}$ is an open covering of X. Taking a finite subcovering of this extended covering yields a finite subcovering of the original covering.

For the second statement, suppose that $\{U_i\}$ is an open covering of $X = \mathrm{Spec}\, A$. For each i, let J_i be the ideal such that $\complement_X U_i = V(J_i)$. We see that $\emptyset = \bigcap V(J_i) = V(\sum J_i)$, so there is a finite family of $f_k \in J_k$ such that $\sum f_k = 1$. Then $\sum J_k = 1$ and $\{U_k\}$ is a finite subcovering. □

The next exercise is a bit open ended in character, so the proof label is changed to reflect that.

Exercise 2.1.7. Describe the closed subsets of $\operatorname{Spec}(\mathbf{Z}[T])$. Show that they are not all principal.

Answer. The closed subsets of $\operatorname{Spec}(\mathbf{Z}[T])$ can be broken up into three families: points, lines, and the generic point. The points are simply maximal ideals; in $\mathbf{Z}[T]$ these are ideals of the form (p, P), where p is a non-zero prime number and P is an irreducible polynomial. The lines are just principal ideals; I call them lines because they determine a family of points that looks kind of like a line. Finally, there is the generic point (0). The fact that the closed subsets are not all principal follows from the fact that the maximal ideals have 2 generators. □

Exercise 2.1.8. Let $\varphi : A \to B$ be an integral homomorphism.

(a) The map $\operatorname{Spec}\varphi : \operatorname{Spec} B \to \operatorname{Spec} A$ sends closed points to closed points, and any point in the preimage of a closed point is also a closed point.

(b) Let $\mathfrak{p} \in \operatorname{Spec} A$. The canonical homomorphism $A_\mathfrak{p} \to B \otimes_A A_\mathfrak{p}$ is integral.

(c) Let $T = \varphi(A \setminus \mathfrak{p})$ and suppose that φ is injective. Then T is a multiplicative subset of B and $B \otimes_A A_\mathfrak{p} = T^{-1}B \neq 0$. In particular, $\operatorname{Spec}\varphi$ is surjective if φ is integral and injective.

Proof. For part (a), suppose that $\mathfrak{m}$ is a closed point of $\operatorname{Spec} B$ and let $\varphi^{-1}(\mathfrak{m}) = \mathfrak{m}'$. Then we have an integral homomorphism $A/\mathfrak{m}' \hookrightarrow B/\mathfrak{m}$. Given $a \in A/\mathfrak{m}'$, the inverse $b \in B/\mathfrak{m}$ of its image under φ satisfies an integral equation $a_0 + a_1 b + \ldots + b^n = 0$. Multiplying by an appropriate power of a, we have $a_0 a^n + \ldots + 1 = 0$, so a is a unit. On the other hand, let $\mathfrak{m}$ be a closed point of $\operatorname{Spec} A$ and suppose that $\varphi(\mathfrak{m}) \subseteq \mathfrak{m}'$ for some $\mathfrak{m}' \in \operatorname{Spec} B$. Again, we have an integral homomorphism $A/\mathfrak{m} \hookrightarrow B/\mathfrak{m}'$. Since $A/\mathfrak{m}$ is a field, this is an algebraic extension.

Parts (b) and (c) are easy, but somewhat annoying to write out. □

Exercise 2.1.9. Let k be a field and A be a finitely generated k-algebra.

(a) Suppose in addition that A is a finite over k. Then $\operatorname{Spec} A$ is a finite set, with cardinality bounded above by $\dim_k A$. Furthermore, every prime ideal of A is maximal.

(b) The cardinality of $\operatorname{Spec}(k[T_1, \ldots, T_d])$ is infinite when $d \geq 1$.

(c) The set $\operatorname{Spec} A$ is finite iff A is finite over k.

Proof. For part (a), the second statement is obvious. To prove the first one, let $\{a_1, \ldots, a_n\}$ be a basis for A over k. We see that there is a family of maximal ideals $\mathfrak{m}_1, \ldots, \mathfrak{m}_n$, where $\mathfrak{m}_i$ is the largest ideal not containing a_i. If $\mathfrak{m}$ is any other maximal ideal, there is some i such that $a_i \notin \mathfrak{m}$, so $\mathfrak{m} = \mathfrak{m}_i$.

In (b), it suffices to prove it when $d = 1$. In this case, it's just Euclid's proof of the infinitude of primes.

Part (c) follows from parts (a) and (b) after an easy application of Noether normalization. □

2.2 Ringed topological spaces

Let X be a topological space.

Exercise 2.2.1. Let A be a non-trivial abelian group and let $\mathcal{A}_X$ be the constant presheaf of A on X. The associated sheaf $\mathcal{A}_X^\dagger$ is the sheaf of locally constant functions valued in A. In particular, $\mathcal{A}_X$ is a sheaf if and only if every non-empty open subset of X is connected.

Proof. Let $\mathcal{F}$ be the sheaf of locally constant A-valued functions on X and let $\varphi : \mathcal{A}_X \to \mathcal{F}$ be the evident morphism. It is easy to see that the stalks of $\mathcal{F}$ are all A. Let $\mathcal{G}$ be any sheaf and let $\alpha : \mathcal{A}_X \to \mathcal{G}$ be any morphism. If f is a locally constant function on U, we may split U into disjoint open sets $U_a = f^{-1}(a)$ for $a \in f(U)$. We define $\widetilde{\alpha}(U) : \mathcal{F} \to \mathcal{G}$ to be the map sending f to the regluing of the sections $\alpha(U)(a)|_{U_a}$. It is easy to see that this defines a morphism of sheaves and that $\widetilde{\alpha} \circ \varphi = \alpha$. Since φ is surjective, $\widetilde{\alpha}$ is the unique such morphism. □

Exercise 2.2.2. Let $\mathcal{F}$ be a sheaf on X. Let $s, t \in \mathcal{F}(X)$. Then the set of $x \in X$ such that $s_x = t_x$ is open in X.

Proof. If $s_x = t_x$ at some point $x \in X$, there must be a neighborhood U_x of x such that $s_y = t_y$ for all $y \in U_x$. Taking the union of all such U_x gives the result. □

Exercise 2.2.3. Let X be a fixed topological space. Let $\mathcal{F}$ be a presheaf on X. We keep the notation introduced just before Lemma 2.7.

(a) Let $\mathcal{U} = \{U_i\}$ be an open covering of X. Let $\mathcal{F}_{\mathcal{U}}(X) = \operatorname{Ker} d_1$. For any open subset W of X, we define in the same way a group $\mathcal{F}_{\mathcal{U}}(W)$ relative to the covering $\{W \cap U_i\}$ of W. Then $\mathcal{F}_{\mathcal{U}}$ is a presheaf on X and we have a morphism of presheaves $\mathcal{F} \to \mathcal{F}_{\mathcal{U}}$.

(b) Let $\mathcal{V} = \{V_j\}$ be another open covering of X. We define a partial ordering: $\mathcal{V} \preceq \mathcal{U}$ if $\mathcal{V}$ is finer than $\mathcal{U}$. Then there is a canonical homomorphism $\mathcal{F}_{\mathcal{U}}(W) \to \mathcal{F}_{\mathcal{V}}(\mathcal{W})$ for every open subset W if $\mathcal{V} \preceq \mathcal{U}$.

(c) The ordering $\preceq$ makes the set of open coverings of X into a direct system. Let
$$\mathcal{F}'(W) = \varinjlim_{\mathcal{V}} \mathcal{F}_{\mathcal{V}}(W),$$
the direct limit being taken over the open coverings of W. Then $(\mathcal{F}')'$ is the sheaf associated to $\mathcal{F}$.

Proof. Part (a) is clear. The homomorphisms in part (b) are given by the resrictions in the natural way.

Part (c) is quite nasty to write out so I won't do it. □

Exercise 2.2.4. Let $0 \to \mathcal{F}' \to \mathcal{F} \to \mathcal{F}''$ be an exact sequence of sheaves on X. Then the sequence of abelian groups $0 \to \mathcal{F}'(X) \to \mathcal{F}(X) \to \mathcal{F}''(X)$ is exact.

Proof. This follows from the exact sequence induced on stalks. □

Exercise 2.2.5. Fix a sheaf $\mathcal{G}$ on X and a closed point $x_0 \in X$. Define a presheaf $\mathcal{F}$ by $\mathcal{F}(U) = \mathcal{G}(U)$ if $x_0 \notin U$ and $\mathcal{F}(U) = \{s \in \mathcal{G}(U) : s_{x_0} = 0\}$ otherwise. Then $\mathcal{F}$ is a sheaf and $\operatorname{Supp}\mathcal{F} = \operatorname{Supp}\mathcal{G} \setminus \{x_0\}$.

Proof. I think I'm missing something because this seems really immediate □

2.3 Schemes

Exercise 2.3.1. Let X be an open subscheme of an affine scheme Y. The canonical morphism $X \to Y$ corresponds by the map ρ of Proposition 25 to the restriction $\mathcal{O}_Y(Y) \to \mathcal{O}_X(X)$.

Proof. In the case where X is a principal open subset, this is just Lemma 3.7. Otherwise, suppose that $X = \bigcup D(f_i)$ for some family of elements $f_i \in \mathcal{O}_Y(Y)$. The inclusions induce a family of maps $\mathcal{O}_Y(Y) \to \mathcal{O}_X(D(f_i)) = \mathcal{O}_X(X)_{f_i}$ which clearly agree on the intersections. Gluing these gives the restriction. □

Exercise 2.3.2. Let $U = \operatorname{Spec} B$ be an affine open subscheme of $X = \operatorname{Spec} A$. Then the restriction $A \to B$ is flat.

Proof. Since the restriction corresponds to the open immersion $U \hookrightarrow X$, we see that its localization $A_x \to B_x$ is an isomorphism, which is flat, for every $x \in \operatorname{Spec} B$. □

Exercise 2.3.3.

(a) Let F be a subset of an affine scheme $\operatorname{Spec} A$. The closure $\overline{F}$ of F in $\operatorname{Spec} A$ is equal to $V(I)$, where $I = \bigcap_{\mathfrak{p} \in F} \mathfrak{p}$.

(b) Let $\varphi : A \to B$ be a ring homomorphism. Let $f : \operatorname{Spec} B \to \operatorname{Spec} A$ be the morphism of schemes associated to φ. Then $\overline{\operatorname{Im} f} = V(\operatorname{Ker}\varphi)$.

Proof. We already have that $\overline{F}$ must already be of the form $V(I')$ with $I' \subset I$. Since $V(I')$ must be the minimal such set, we have $I' = I$.

For part (b) we assume that A and B are reduced, since that doesn't affect their prime spectra. Then the previous part implies that $\overline{\operatorname{Im} f} = V(\bigcap_{\mathfrak{p} \in \operatorname{Spec} A} \varphi^{-1}(\mathfrak{p})) = V(\operatorname{Ker}\varphi)$. □

In the case where $B = A_{\mathfrak{p}}$ for some $\mathfrak{p} \in \operatorname{Spec} A$, this is the set of ideals that are "large enough" to interact with $\mathfrak{p}$. In geometric terms, they are the irreducible varieties such that every function that vanishes on $V(\mathfrak{p})$ vanishes here as well.

Exercise 2.3.4. Let X be a scheme and $f \in \mathcal{O}_X(X)$. Then the map $U \mapsto f|_U \mathcal{O}_X(U)$ for every affine open subset U defines a sheaf of ideals on X, denoted $f\mathcal{O}_X$. In addition, $\operatorname{Supp} f\mathcal{O}_X$ is closed.

Proof. It is easy to see that $f\mathcal{O}_X$ inherits the uniqueness condition from $\mathcal{O}_X$. On the other hand, suppose that we have an open set U of X, an affine open cover $\{U_i\}$ of U, and a family of sections $f|_{U_i}s_i$ that agrees on intersections. Then we may glue these into a section s of $\mathcal{O}_X(U)$ that restricts to $f_{U_i}s_i$ on every U_i, which we contest is contained in $f\mathcal{O}_X(U)$. Indeed, suppose that s is not in this ideal. Then for any $a \in \mathcal{O}_X(U)$, there is some $x \in U$ such that $s_x - (af)_x \neq 0$, but this contradicts the fact that $f|_{U_i}s_i$ is in $f\mathcal{O}_X(U)$.

To see that $\operatorname{Supp} f\mathcal{O}_X$ is closed, observe that $\operatorname{Supp} f\mathcal{O}_X$ is just the set of $x \in X$ such that $f_x \neq 0$. Since the complement is open, the statement holds. □

Exercise 2.3.5. Let Y be a scheme such that $\operatorname{Mor}(X, Y) \cong \operatorname{Hom}(\mathcal{O}_Y(Y), \mathcal{O}_X(X))$ for any affine scheme X. Then Y is affine.

Proof. Regrettably, the easiest proof I could come up with was by categoryslop. Observe that for any affine scheme Z, we have $\operatorname{Mor}(Z, Y) \cong \operatorname{Hom}(\mathcal{O}_Y(Y), \mathcal{O}_Z(Z)) \cong \operatorname{Mor}(Z, \operatorname{Spec}(\mathcal{O}_Y(Y)))$. This extends to the category of all schemes in the obvious way, so Y is a representing object of $\operatorname{Mor}(-, \operatorname{Spec}(\mathcal{O}_Y(Y))$ and is thus isomorphic to $\operatorname{Spec}(\mathcal{O}_Y(Y))$. □

2.4 Reduced schemes

Exercise 2.4.1. Let k be a field and $P \in k[T_1, \ldots, T_n]$. Then $\operatorname{Spec}(k[T_1, \ldots, T_n]/(P))$ is a reduced (resp. irreducible; integral) if and only if P is squarefree (resp. admits only one irreducible factor; is irreducible).

Proof. dawg □

Exercise 2.4.2. Let X be a scheme and $x \in X$. Then the image of the morphism $\operatorname{Spec} \mathcal{O}_{X,x} \to X$ is the set of points of X that specialize to x.

Proof. It suffices to prove this in the case where X is affine. If y is any point of X that specializes to x, we have $\mathfrak{j}_y \subseteq \mathfrak{j}_x$ and so y is in the image of $\operatorname{Spec} \mathcal{O}_{X,x}$. The other direction is basically the same. □

Exercise 2.4.3. Let $\mathcal{O}_K$ be a discrete valuation ring with field of fractions K and uniformizing parameter t. Then $\operatorname{Spec} K[T]$ embeds as an open subscheme of $\operatorname{Spec} \mathcal{O}_K[T]$.

Proof. Observe that $(\mathcal{O}_K)_t = K$, so $(\mathcal{O}_K[T])_t = K[T]$. Since $\operatorname{Spec}((\mathcal{O}_K[T])_t)$ is homeomorphic to $D(t) \subseteq \operatorname{Spec}(\mathcal{O}_K[T])$, this proves the proposition. □

The second part of this exercise is to find the points of $\operatorname{Spec} K[T]$ which specialize to the point corresponding to the maximal ideal (T, t) of $\mathcal{O}_K[T]$. To do this, notice that $\operatorname{Spec} K[T]$ corresponds to the ideals I of $\mathcal{O}_K[T]$ such that $I \cap (t) = \{0\}$, i.e. the principal ideals generated by nonconstant polynomials. Then the only points that specialize to (T, t) are (0) and (T).

Exercise 2.4.4. Let X be a scheme having only a finite number of irreducible components $\{X_i\}$. Then X is connected if and only if for any pair i, j, there exist indices $i_0 = i, i_1, \dots, i_r = j$ such that $X_{i_l} = X_{i_{l+1}}$ for every $l < r$. Furthermore, X is integral if and only if it is connected and integral at every point.

Proof. Since the irreducible components of X are finite in number, they must all be open; it is clear that having X connected implies the condition. On the other hand, suppose that X has the property. If U is a nonvoid open set such that $\overline{U} = U$, we have that U i the union of irreducible components of X, so suppose that X_i is any component (not neccessarily contained in U). Then we have a sequence $X = X_0, \dots, X_r \subseteq U$ such that $X_i \cap X_{i+1} \neq \emptyset$. However, since $X_{r-1} \cap U \neq \emptyset$, X_{r-1} is contained in U and so we may drop the last index. Repeating this, we may assume that $r = 0$ and so $X \subseteq U$.

Now, if X is integral, it is obvious that it is connected and integral at every point, so assume the latter. At any point $x \in X$ let $i : \operatorname{Spec} O_{X,x} \to X$ be the canonical morphism. Since $i^{-1}(i((0))) \subseteq i^{-1}(\overline{i((0))})$, we have that $\operatorname{Spec} O_{X,x} \subseteq i^{-1}(\overline{i((0))})$ and so the image of i has a closed point. It follows from exercise 2 that each point x is contained in exactly one irreducible component, so the connectedness hypothesis implies that X is irreducible. That X is reduced is obvious, so X is integral. □

Exercise 2.4.5. Let X be a scheme. Every irreducible component of X is contained in a connected component. If X is locally noetherian, then the connected components are open. If X is noetherian, the connected components are finite in number.

Proof. The first and third statements are clear, and the second one follows from the third. □